

# SSQ-LR: Sub-FP16 Recurrent State Quantization for Hybrid Reasoners

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## Abstract

This is a living paper shell for a preregistered Mac-first experiment. SSQ-LR tests whether recurrent SSM state in hybrid reasoning models can be quantized below FP16 while preserving quality. No positive result or native systems claim is made yet; current S2/S3 scouts stop the frozen recipe, and no recipe clears quality, byte accounting, and no-retuning transfer.

## 1 Current Gate and Claim Boundary

The branch must pass state-distribution heterogeneity, state-only quantization sensitivity, and cross-model transfer gates before any GPU validation. S1 must first show real state heterogeneity on a live hybrid model; S2 then freezes one state-only quantization recipe; S3 tests that recipe without per-model retuning. Failing any gate kills the branch before GPU work. The current real trace packet builder/checker covers S1. S2 and S3 now have executable follow-up contracts in `experimental/shared/followup_gate_contracts.py`: they require frozen recipe IDs, byte accounting, BF16 no-op controls, paired accuracy and continuation-NLL quality bounds, verbosity/length-drift readouts before revival, frozen recipe/source hashes, and no-retuning transfer rows. The byte gate is counted as

$$\frac{\text{BF16 state bytes}}{\text{quantized state bytes} + \text{scale bytes} + \text{metadata bytes}'}$$

so scale tensors, layer masks, and recipe metadata are not free. Resource-limited S2 scouts are not current evidence for promotion. SSQ-LR makes no quality, byte-savings, transfer, GPU, latency, or throughput claim until non-resource-limited S1–S3 packets clear.

## 2 Mac Artifact Status

A deterministic synthetic S1 packet now validates the real-schema checker path before real hybrid-model state dumps. It emits 288 rows covering 12 prompts, 6 recurrent layers, and the four preregistered buckets, producing `SCHEMA_REHEARSAL_NOT_PROMOTABLE_SYNTHETIC_SSQ_LR_S1`. This is synthetic-only: it is not model evidence, not a quality result, and not GPU evidence. The held-out S1b packet is the strongest positive evidence so far: it repeats the frozen selected layers 0, 12, and 30 on 12 held-out prompts with selected ratio 2.459 and lower bound 1.861. Uniform state recipes do not yet convert that heterogeneity into a method. MXFP4 preserves short BF16-argmax fidelity but reaches only  $3.765\text{--}3.938\times$  state-memory reduction after scale bytes. INT3 reaches  $4.923\text{--}5.224\times$  after scale bytes, but both 12-prompt held-out scouts fail because at least one prompt loses argmax fidelity. The 4-prompt INT3 pass is therefore treated as resource-limited and non-promoting. The newest resource-limited mixed-block S2b scout selects `mixed.int3.mxfp4.low_error_25pct`: it reaches  $4.192\times$  counted state-memory reduction on the 12-prompt held-out replay with zero BF16-argmax delta and 0.03956 selected NLL-delta CI high. This short-window pass is still simulated replay and cannot promote to GPU. The same recipe then fails a longer-window replay (`--max-input-tokens 24 --prefix-tokens 8`): it preserves the  $4.192\times$  byte readout but has selected accuracy CI high 0.0667 and selected NLL-delta CI high 0.0764. The mixed recipe is therefore stopped before GPU validation. Layer-localization gives one narrower layer

set: layers 0 and 30 individually pass the same longer-window replay with INT3 state at  $5.224\times$  counted memory reduction and zero selected accuracy delta, while layer 12 fails. The combined layers 0 and 30 replay also passes with selected NLL-delta CI high 0.04294. A stricter prefilter replay on this layer set weakens pure INT3 (accuracy CI high 0.105) but selects `mixed_int3_mxfp4_low_error_25pct` at  $4.192\times$  counted memory, zero selected accuracy drift, and 0.05044 selected NLL-delta CI high. The S3 transfer replay then fails on the second complete local hybrid model: Granite 350M with the same 12 prompts, layers 0 and 30, 32-token context, and no retuning selects an INT8 fallback at only  $1.984\times$  memory. Layer localization is post-hoc diagnostic evidence only: layer 0 can pass on Granite 350M and layer 30 fails, but the source Granite Tiny layer-0 replay and Granite 350M layer-0 replay prefer different low-bit recipes. Two local S3 packets that freeze either `mixed25` or INT3 for layer 0 both fail, so there is no current frozen recipe that transfers.

| Mac packet                                        | Selected recipe                 | Memory        | Accuracy CI high |
|---------------------------------------------------|---------------------------------|---------------|------------------|
| Granite Tiny layer 0 source                       | <code>mixed25 INT3/MXFP4</code> | $4.192\times$ | 0.000            |
| Granite 350M layers 0,30                          | INT8 fallback                   | $1.984\times$ | 0.000            |
| Granite 350M layer 0                              | INT3                            | $5.224\times$ | 0.000            |
| Granite 350M layer 30                             | INT8 fallback                   | $1.984\times$ | 0.000            |
| Two-model S3, frozen <code>mixed25</code> layer 0 | <code>mixed25 INT3/MXFP4</code> | –             | 0.0667           |
| Two-model S3, frozen INT3 layer 0                 | INT3                            | –             | 0.0526           |

Table 1: Current SSQ-LR transfer failure. Layer diagnostics are not a rescue: the recipe that passes one Granite model does not pass the other under the frozen no-retuning S3 contract. The rows are Mac-local simulated replay, not native GPU, bandwidth, or serving evidence.

### 3 Next Evidence

The next admissible action is not GPU validation. SSQ-LR needs a new preregistered recipe or layer-selection rule that is frozen before seeing transfer rows, then must rerun no-retuning transfer plus verbosity/length-drift readouts. The current S1b/S2b result is diagnostic evidence, not a positive method.

### 4 Admissible Real Packet

Promotable packets must pin model/tokenizer revisions, prompt and trace-plan hashes, the shared architecture-map hash, exact command/device/dtype metadata, and saved recurrent-state tensor hashes. S1 rows must cover all four registered position buckets, including `prefill_end`, for every prompt/layer pair and at least 12 prompt IDs unless explicitly marked resource-limited. The checker recomputes row metrics and gate aggregates from raw tensors, rejects stale summaries, and requires `RESOURCE_LIMITED_NOT_PROMOTABLE` decisions for any subset packet. Full schema details live in the README and runbook.

### 5 Reviewer Packet

The reviewer-facing claim boundary is tracked in `experimental/ssq.lr/paper/reviewer-pack.md`. It records that SSQ-LR is not camera-ready as a method paper until non-resource-limited, passing S1–S3 evidence exists. The newest novelty baseline is Nemotron 3 Super’s recurrent-cache recipe: NVIDIA documentation treats Mamba SSM cache quantization as a separate deployment challenge and selects FP16 with stochastic rounding for the SSM cache, with INT16/block-scaled variants discussed as controls. SSQ-LR is therefore not novel for noticing SSM cache precision; it must beat this baseline with a sub-FP16 recipe and paired quality evidence. The current local recipes do not beat this baseline. The source memo is `../../../../references/754_ssq_lr_nemotron_state_cache_refs_20260507.md`.

| Required baseline                  | Purpose                                                               |
|------------------------------------|-----------------------------------------------------------------------|
| BF16 no-op replay                  | Catch hook/serialization bugs.                                        |
| FP16 stochastic-rounding SSM cache | Deployment baseline from Nemotron-style hybrid serving.               |
| INT16 block-scaled SSM cache       | Higher-precision recurrent-cache control.                             |
| INT8 state-only                    | Standard low-bit state control.                                       |
| FP8-style state-only               | Checks whether sub-FP16 state can match the likely deployed fallback. |
| MXFP4 state-only                   | Candidate sub-FP16 recipe.                                            |
| INT3 state-only                    | Tests whether sub-4-bit state can clear scale-byte overhead.          |
| Random same-L2 noise               | Separate structure from noise magnitude.                              |
| Scale shuffle                      | Test scale assignment sensitivity.                                    |
| Byte accounting                    | Include scale/metadata overhead.                                      |

Table 2: Controls required before SSQ-LR can claim a state-quantization recipe.

## 6 Claim Boundary

SSQ-LR is only about recurrent SSM-state precision in hybrid reasoners. It does not claim weight quantization, activation quantization, KV-cache quantization, native serving speed, or memory savings until a real state recipe and later GPU validation pass their preregistered gates.